Student 13 **Gudi Yashwanth** 192524275RD 10 / 10 submitted

Q1. Sum the given number upto single digit

CODE

```
n = input().strip()

while len(n) > 1:
    n = str(sum(int(d) for d in n))

print(n)
```

INPUT 89789

OUTPUT (no output)

Q2. Prime Number

CODE

```
def is_prime(n):
    if n < 2:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True

n = int(input())
n += 1
while not is_prime(n):
    n += 1

print(n)
```

INPUT 13

OUTPUT (no output)

Q3. Swap the first and last digits

CODE

```
n = input()
if len(n) == 1:
    print(n)
else:
    swapped = n[-1] + n[1:-1] + n[0]
    print(swapped)
```

INPUT 23546

OUTPUT (no output)

Q4. Odd Sequence first and Even Sequence next

CODE

```
n = input()
odd_digits = ""
even_digits = ""

for ch in n:
    if int(ch) % 2 != 0:
        odd_digits += ch
    else:
        even_digits += ch

print(odd_digits + even_digits)
```

INPUT 658134

OUTPUT (no output)

Q5. Step Number

CODE

```
n = input().strip()
num = int(n)
if len(n) == 1:
    print("Yes")
else:
    ok = True
    for i in range(len(n) - 1):
        if abs(int(n[i]) - int(n[i+1])) != 1:
            ok = False
            break
    print("Yes" if ok else "No")
```

INPUT 123454321

OUTPUT (no output)

Q6. Reverse only the even numbers

CODE

```
s = input().strip()
digits = list(s)

even_positions = [i for i, ch in enumerate(digits) if int(ch) % 2 == 0]
even_digits = [digits[i] for i in even_positions]

even_digits.reverse()

for idx, pos in enumerate(even_positions):
    digits[pos] = even_digits[idx]

print("".join(digits))
```

INPUT 325649

OUTPUT (no output)

Q7. Encoding

CODE

```
n = input().strip()
count = 0
prev = None
encoded = ""
for ch in n:
    if ch ≠ prev:
        count = 1
        prev = ch
    else:
        count += 1
groups = []
prev = n[0]
cnt = 1
for i in range(1, len(n)):
    if n[i] == prev:
        cnt += 1
    else:
        groups.append(prev)
        groups.append(str(cnt))
        prev = n[i]
        cnt = 1
groups.append(prev)
groups.append(str(cnt))
encoded = "".join(groups)
print(encoded)
```

INPUT

```
55553322255
```

OUTPUT

```
(no output)
```

Q8. Reverse Odd Digit

CODE

```
s = input().strip()
odd_idxs = []
odd_digits = []
for i, ch in enumerate(s):
    if int(ch) % 2 ≠ 0:   # odd digit
        odd_idxs.append(i)
        odd_digits.append(ch)
odd_digits.reverse()
arr = list(s)
for k, idx in enumerate(odd_idxs):
    arr[idx] = odd_digits[k]
print(''.join(arr))
```

INPUT

```
816253
```

OUTPUT

```
(no output)
```

Q9. Convert every number into the next sequence number.

CODE

```
s = input().strip()

out = ""
for ch in s:
    if ch.isdigit():
        out += str((int(ch) + 1) % 10)   # next digit (9 → 0)
    else:
        out += ch

print(out)
```

INPUT

```
89781
```

OUTPUT

```
(no output)
```

Q10. Print the total factor count of a given number

Score: 5.00 **TC: 1/1 passed**

CODE

```
n = int(input())

count = 0
for i in range(1, n + 1):
    if n % i == 0:
        count += 1

print(count)
```

INPUT

```
12
```

OUTPUT

```
6
```